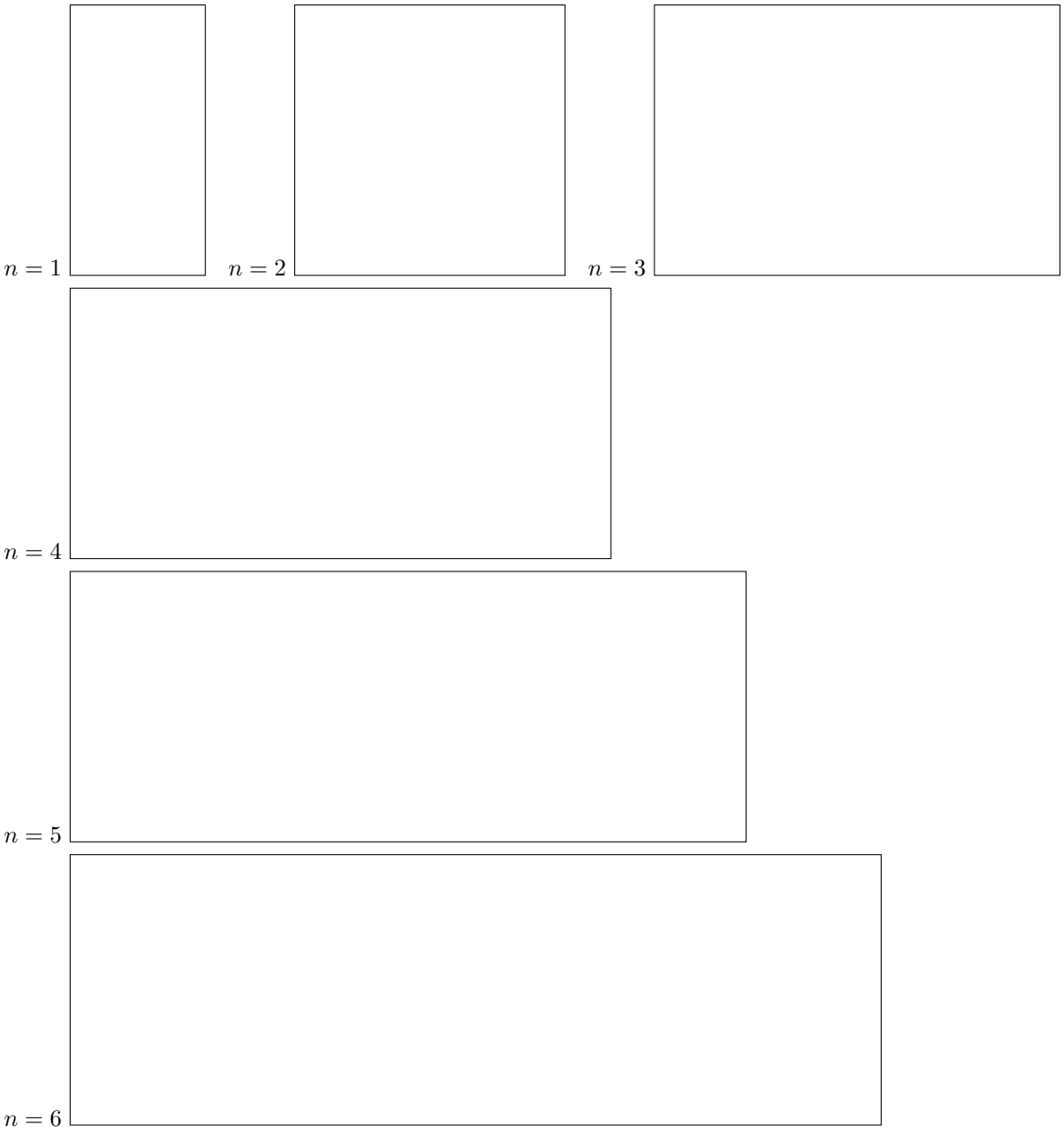
Complements of Kappa Mu Epsilon
(Math Club)

Instructions: The goal is to fill in the numbers **1** through **81** in the grid so that the numbers increase in the snake where the snake can only go up, down, left, or right. For example, the blank box in the upper left corner must be a 51 because 51 has to be next to 50 and the 50 is stuck with only one blank box next adjacent.

49	48	47	46	45	4	5	6	7
50								8
81								13
76								14
75								19
66								20
65								21
64								28
63	62	61	60	33	32	31	30	29

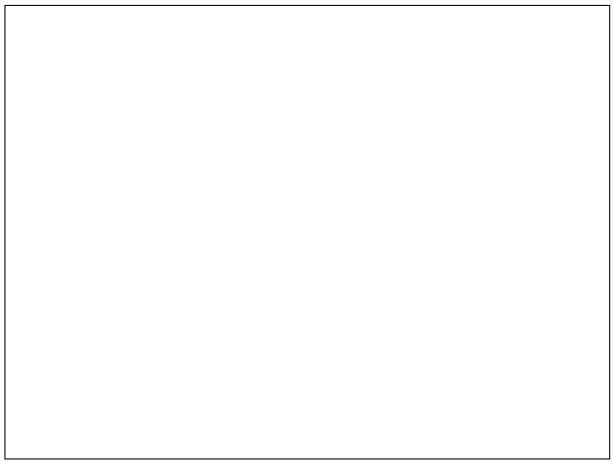
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1. How many different ways are there to place the dominoes in a $2 \times n$ hallway?



2. What would be a formula for a general n? Explain why the pattern and formula work.

3. There are three ways to dissect a 3×2 rectangle into dominos. How many ways are there to dissect a 3×4 rectangle into dominos?



4. It is not possible to dissect a 3×1 rectangle into dominos. Likewise it is not possible to dissect $3 \times 33 \times 5, 3 \times 7, \ldots$ rectangles into dominos. Suppose we remove a single 1×1 square from the lower left corner of these rectangles (we'll call these shapes the $3 \times 1 - 1, 3 \times 3 - 1, 3 \times 5 - 1, \ldots$). These shapes can be dissected into dominos. Below we show, for example, how to dissect the $3 \times 1 - 1$ shape, $3 \times 3 - 1$ shape, and $3 \times 5 - 1$ shape into dominos. How many ways are there to dissect a $3 \times 3 - 1$ rectangle into dominos? A $3 \times 5 - 1$ rectangle into dominos?

